GEK-99645

Instructions

IC5281A DC Contactor

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CHAPTER 1

GENERAL INFORMATION

1-1. SAFETY PRECAUTIONS

The IC5281A dc contactor is installed in equipment that uses hazardous voltages. This manual includes warning and caution statements that must be observed to ensure the safety of personnel and prevent equipment damage. It is important to fully understand these warnings and cautions and the instructions before maintaining this equipment.

1-2. WARNING STATEMENTS. A warning statement alerts personnel to a step of a procedure that, if not strictly followed, could seriously injure or kill someone.

1-3. CAUTION STATEMENTS. A caution statement alerts personnel to a step of a procedure that, if not strictly followed, could damage or destroy equipment.

1-4. INTRODUCTION AND DESCRIPTION

This manual provides installation, operating, and maintenance instructions for GE's IC5281A dc contactor. This contactor is identified by the complete part number, located on its nameplate:

IC5281A108A2RVNPG1

This part number exactly identifies the equipment when ordering replacement parts.

GE drawing 238B2713 defines this contactor for engineering applications. (Engineering drawings are not included in this manual.)

1-5. EQUIPMENT SPECIFICATIONS. The contactor meets the requirements of Military Specification MIL-C-2212 for Naval Shipboard use, except for interruption endurance.

Table 1. Interruption Endurance With 0.2 Millihenries Inductance

No. Interruptions	Electrical Rating
49.950	500 A at 50 V dc
10	3500 A at 355 V dc
20	3500 A at 270 V dc
20	1675 A at 355 V dc
4.000	1300 A at 270 V dc

The contactor is designed for shipboard application with the characteristics listed below:

Rating: 250 V dc (180 V dc to 355 V dc), 1500 A

Insulation system: Class F (155 °C)

Ambient: 70 °C at contactor

Weight: 70 lbs. for contactor
16 lbs. repair parts (unboxed)

Shock: Grade A, Class 1, Type C
in accordance with MIL-S-901

CHAPTER 2

PREVENTIVE MAINTENANCE

2-1. INTRODUCTION

Preventive maintenance consists of inspections, tests, and cleaning of equipment at scheduled intervals. It helps detect and correct conditions that could cause equipment malfunction.

2-2. SAFETY PRECAUTIONS

WARNING

Disconnect all power supplies before performing any maintenance, adjustments, servicing, parts replacements, or any other act requiring physical contact with the electrical working components or wiring of this equipment.

This contactor is installed in systems that use high voltages. If the contactor is serviced while installed in such a system, it is important that only experienced and authorized technicians who observe standard safety precautions perform all phases of installation, operation, and maintenance. Before conducting any preventive maintenance, fully deenergize and ground the equipment. Tag the equipment Out of Service only after it has been fully deenergized.

2-3. SERVICING LIVE EQUIPMENT. If it is necessary to service live equipment, the following general instructions must be followed:

WARNING

Use only one hand when servicing live equipment. This prevents an accidental current path being created through the body from one hand to the other.

a. Deenergize the equipment containing the contactor. To deenergize any capacitors connected to these terminals, temporarily ground the terminals to be measured.

b. Connect the multimeter to the terminals to be measured, using a range higher than the expected voltage. Make sure that you are not grounded whenever you adjust equipment or use measuring equipment. Do not use test equipment known to be in poor condition.

c. Without touching the meter or test leads, energize the equipment and read the meter.

d. Remove the test leads after deenergizing the equipment.

2-4. PREVENTIVE MAINTENANCE SCHEDULE

The schedule for conducting preventive maintenance depends on how much the contactor is used. Table 2-1 lists the suggested maintenance schedule, which is based on average conditions. This schedule should be adjusted for each application as experience requires. If the host equipment (the system in which the contactor is mounted) has it's own schedule, the user should disregard Table 2-1.

Table 2-1. Preventive Maintenance Schedule

Frequency of equipment operation	Maintenance interval
Four or more operations per hour	Monthly
Automatic operation	Quarterly
Non-automatic operation (including propulsion systems)	Semi-annually
All others	Annually

2-5. CONTACTORS. Check contactors as follows:

a. Operate contactors by hand to check that linkages operate freely from mechanical binding in the normal operating position. When released from the energized position, the contactor must drop out to the full deenergized position. If there is a question about these positions, first energize the contactors electrically, then deenergize them and check that the contactor drops out to the full deenergized position.Sticky operation may be improved by cleaning the parts with Freon-TF then applying a light coating of grease to all bearing points.

c. Inspect the ac magnet structures for dirty or rusted mating surfaces, uneven wear or pitting of mating faces, broken or missing pole shaders, or loose laminations from damaged or sheared rivets. An abnormally noisy contactor often indicates these conditions.

d. Inspect the coil for signs of overheating, as indicated by discoloration of the normally green coil.

e. Examine the auxiliary contacts for evidence of excessive burning and pitting. If necessary, replace them (see figure 2-1).

f. Examine the auxiliary contact for alignment. Check the alignment of any pickup and IC5181 contactors, and of any IC5182 ac relays. If required, adjust as follows (see figure 2-1):

(1) Loosen the screws holding the housing.
(2) Realign the housing.
(3) Retighten the screws.

g. Examine the main contacts for signs of excessive burning or pitting. If necessary, replace them.

h. Check for contact wear, as described below.

2-6. Contact Wear. Contact wipe, or wear allowance, is the amount the movable contact support travels after the contacts touch. Contact wipe should be maintained at .100 ±.015 inch. Measure the main contacts for wear allowance as follows (see figures 2-2 and 2-3):

a. Remove the contactor top (x) by removing the screws (x).

b. Connect a light or bell set across a set of stationary contacts (x).

c. Move the magnet towards the closed position until the tips just touch (indicated by the light or bell set), then measure the position of the movable contact carrier (x) with reference to a fixed point on the contactor.

Figure 2-1. Auxiliary Contact Alignment, IC5281 Devices

d. Move the magnet to the fully closed position, then measure the position of the movable contact carrier, as above (step c).

e. Subtract the first value (step c) from the second value (step d) to get the contact wipe.

f. If contact wipe is still out of range, adjust the contact wipe as follows:

(1) Loosen the two screws (one on each side of the return spring) that hold the movable contact carrier.
(2) Move the movable contact carrier up or down to the correct contact wipe.
(3) Check that all tips have the same contact wipe within ±.015 inch.
(4) Tighten the screws and replace the top.
(5) Check for freedom of operation.

Figure 2-2. Contactor, Exploded View

Figure 2-3. Auxiliary Contacts, Exploded View

CHAPTER 3

FUNCTIONAL DESCRIPTION

3-1. CONTACTOR OPERATION

The IC5281A contactor is operated by a shunt-coil magnet. The magnet plunger connects to a lever that brings the movable tip structure forward when the coil is energized. This closes the main and auxiliary contacts. The contactor normally contains two auxiliary contact units, one on each side of the contactor. Each of these units has two double break contacts. Each set of stationary contacts in the auxiliary contact units can be assembled as normally open or normally closed.

3-2. THEORY OF OPERATION

Refer to Figure 3-1.

3-3. CLOSING SEQUENCE. Control power applied to terminals 17 and 18 energizes the coil at full voltage (9.6 A at 250 V dc). Near the end of armature travel, the normally closed auxiliary contact (M) opens without arcing. This transfers coil current to capacitors C1 and C2 through resistor RS. Current then transfers to resitor RD, resulting in reduced coil watts and holding current of 0.4 A at 250 V dc.

The coil filter (diode D and resistor RF), suppresses the voltage transient in capacitors C1 and C2. This reduces the magnetic flux decay rate, which ensures positive magnetic sealing.

3-4. OPENING SEQUENCE. The coil circuit is interrupted externally (outside the contactor). Coil filters (diode D and resistor RF) reduce voltage transients generated by the coil. Capacitors C1 and C2 discharge through resistor RS.

3-5. POWER CIRCUIT. Three parallel contacts carry current. The outboard two contacts carry 98% of the current. The center contact carries 2% of the current because of the blowout current resistance when the contactor opens.

Th outboard contacts open first without arcing and transfer all current to the center contact and blowout coils. The center contact then opens. The resulting arcs are propelled off the contacts and into the arc chute by the blowout coil's field. Interruption completes as the arc attaches to a series of steel splitter plates.

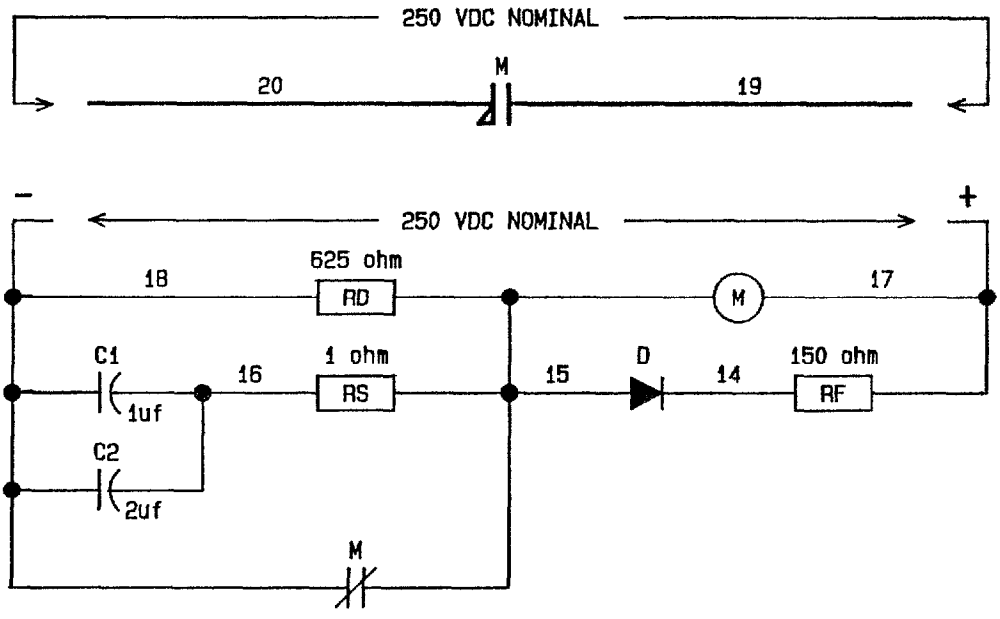


Figure 3-1. IC5281A Contactor Schematic Diagram

CHAPTER 4
TROUBLESHOOTING

4-1. INTRODUCTION

This chapter contains instructions to locate and correct problems in the contactor. (Chapter 5 provides corrective maintenance procedures.) Before troubleshooting the equipment, personnel should be thoroughly familiar with chapters 1 thru 3 of this manual. All warning and caution statements in the text and on equipment near the contactor must be strictly followed.

4-2. TROUBLESHOOTING PROCEDURES

WARNING

Before any maintenance, servicing, adjustments, parts replacements, or any other act is performed requiring physical contact with the electrical working components or wiring of this equipment, disconnect all power supplies, then ground and discharge the equipment.The following paragraphs and tables provide specific troubleshooting guidelines.

Table 4-1 is a trouble analysis chart for the contactor.

Table 4-1. Contactor Trouble Analysis Chart

Symptoms	Probable cause	Remedy
CONTACTS		
Chatter	Poor contact in control	Check contact wipe. Adjust if needed. Replace worn contacts.
	Low voltage	Correct voltage source.
	Accumulated dirt or grease	Clean with dry rag.
	Misalignment	Adjust.
Overheating	Insufficient contact pressure	Adjust contact wipe. Replace contacts if worn.
	Overload	Reduce load current.
Short life	Excessive filing or dressing	Replace contact(s). File only to remove projections above main contact surface.
	Overload	Reduce load current.
	Excessive jogging	Reduce jogging frequency.
	Loose connections	Tighten.
	Oxidation	Clean or replace contact(s).
COILS		
Open circuit	Mechanical damage	Replace coil.
Overheated coil	Wrong coil	Check coil ratings. Replace with correct coil.
	Excessive jogging	Reduce jogging frequency.
	Overvoltage source	Correct voltage source.
	Improper frequency	Correct frequency.

Table 4-1. Contactor Trouble Analysis Chart – Continued

Symptoms	Probable cause	Remedy
MAGNETS, MECHANICAL PARTS		
Wrong or broken part	Heavy slamming caused by:	
	Overvoltage	Correct voltage source.
	Wrong coil	Replace with correct coil.
	Chattering caused by broken pole shader or too much jogging.	Replace part and correct cause of damage.
Noisy magnet	Broken pole shader, or magnet faces not true as a result of wear or mounting strains	Adjust or replace
	Dirty magnet faces	Clean magnet faces.
Failure to pick up	Low voltage on coil	Correct voltage source.
	Open wiring of coil or shorted turns	Replace coil.
	Wrong coil	Check coil ratings. Replace with correct coil.
	Open fuse or circuit breaker	Check fuses & circuit breakers. Determine cause of open device.
	Open overload relay	Check overload relay contacts.
	Open connection	Check control circuit wiring.
Contactor picks up but does not seal	Defective auxiliary contact	Replace auxiliary contact.
Contactor won't drop out	Control short circuit	Remove short circuit.
	Magnet faces coated with sticky substance	Clean magnet faces.

4-3. CORRECTIVE MAINTENANCE

Corrective maintenance is required when parts need adjustment, alignment, repair, or replacement to correct equipment trouble. Under normal service conditions, many of the components will not require repair or replacement.

WARNING

Disconnect all power supplies before performing any maintenance, adjustments, servicing, parts replacements, or any other act requiring physical contact with the electrical components or wiring of this equipment.

4-4. ADJUSTMENTS AND ALIGNMENTS

4-5. Coil Replacement. Replace a contactor coil as follows (refer to figure x):

To be provided.

4-6. Return Spring Replacement. Replace a return spring (7) as follows (refer to figure x):

To be provided.

4-7. Contact Assembly. The contact assembly contains the movable and stationary contact tips and the tip spring. Replace one or all of the parts as follows (refer to figure x):

To be provided.

4-8. Auxiliary Contacts. Replace the auxiliary contacts as follows (refer to figures x):

To be provided.

CHAPTER 5

PARTS LIST

5-1. INTRODUCTION

This chapter identifies the major components of the GE IC5281A dc contactor. Table 5-1 lists the parts and figures 5-1 and 5-2 show the the parts. These enables the user to identify and order replaceable parts.

5-2. USING THE PARTS LIST. In table 5-1, the first column contains references the drawing and the item number in that drawing for each part. The FSCM column identifies the part manufacturer's Federal Supply Code identification number. The other columns describe the part, list the number of parts per controller, and identify the manufacturer's part number for ordering that part. To use the table for replacing a part:

a. In the figures in this chapter, locate the item to be replaced and note the item identification number in the figure. Then refer back to table 5-1.

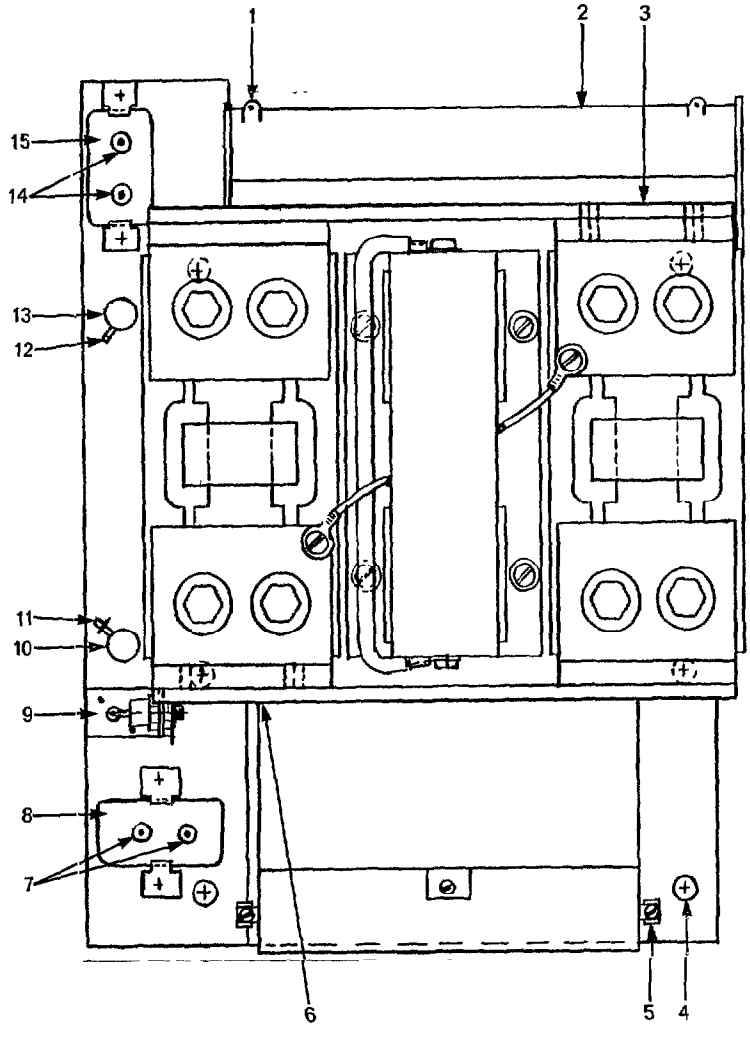
b. Match the item number in the figure with that listed in the table. Then go to the column that lists the manufacturer's part number for that item.

c. Find the FSCM number for the manufacturer. Then go to table 5-2, which identifies the manufacturer by the FSCM number and gives the manufacturer's address.

d. Contact the manufacturer and order the number of parts needed, using the manufacturer's part number and the description of the part.

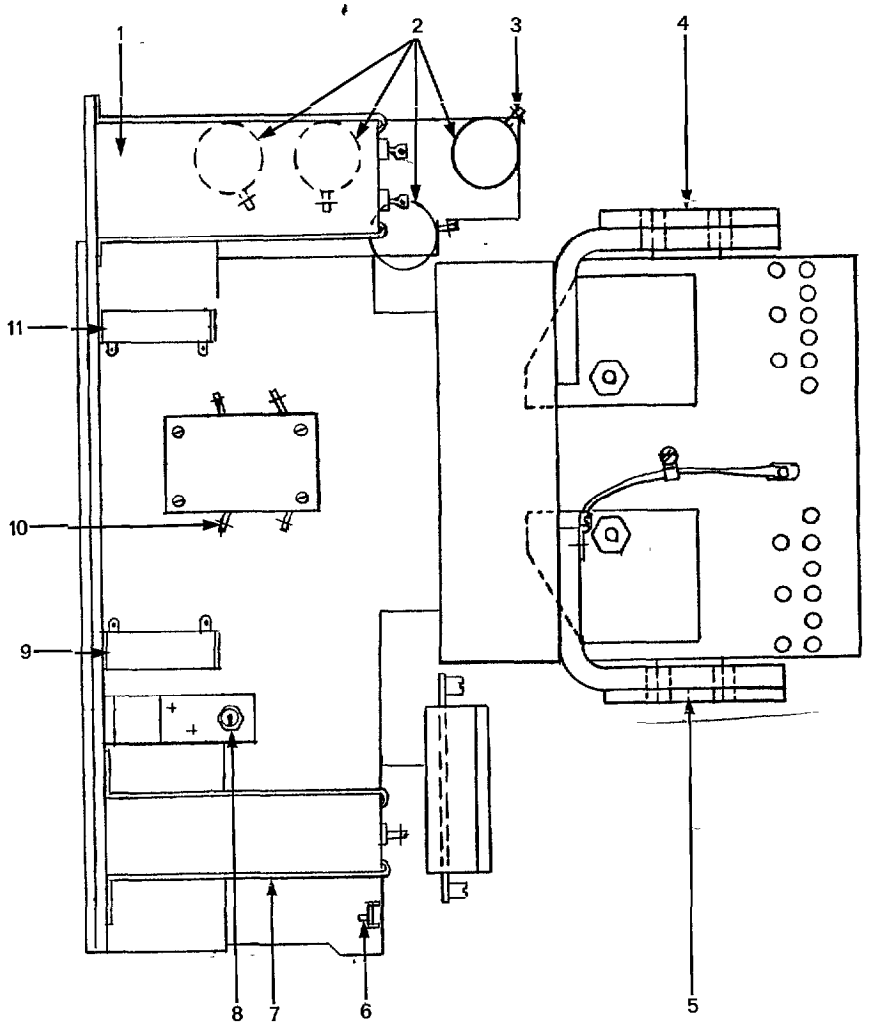
Table 5-1. Contactor IC5281A Parts List

Figure and item no.	Name of part	Quantity per set	Manufacturer's part no.	FSCM
7-x,	Coil, 250 V dc	1	277A9705G2	02989
7-x,	Movable contact, main tip	2	238B2412G1	02989
7-x,	Movable contact, arcing tip	1	238B2421G1	02989
7-x,	Stationary contact, main tip	2	238B2410G1	02989
7-x,	Stationary contact, arcing tip	2	238B2410G1	02989
7-x,	Contact spring, main tip	4	277A9781P1	02989
7-x,	Contact spring, arcing tip	2	277A9782P1	02989
7-x,	Return spring	2	193A9474P1	02989
7-x,	Auxiliary contact, movable assembly	3	129B6882G5	02989
7-x,	Auxiliary contact, movable assembly	1	129B6282G6	02989
7-x,	Auxiliary contact, stationary assembly	14	174A7137P1	02989
7-x,	Auxiliary contact, stationary assembly	2	174A7137P2	02989
7-1, 13 7-2, 11	Resistor (RS)	1	68A9725P2	44655 & 63743
7-1, 2 7-2, 2	Resistor (RD)	4	68A9725P5	44655 & 63743
7-1, 10 7-2, 9	Resistor (RF)	1	68A9725P6	44655 & 63743
7-1, 9 7-2, 8	Diode (D)	1	JANTX-1N3673A	02989
7-1, 8 7-2, 7	Capacitor, (C1)	1	CQ72B1KG105K1	02989
7-1, 15 7-2, 1	Capacitor, (C2)	1	CQ72B1KG205K1	02989



1 Terminal 18
2 Resistors RD
3 Terminal 20
4 Hole for mounting bolts
5 Terminal B
6 Terminal 19
7 Solder connections
8 Capacitor C1
9 Diode D
10 Resistor RF
11 #8 Terminal screws
12 #8 Terminal screws
13 Resistor RS
14 Solder connections
15 Capacitor C2

Figure 5-1. Contactor Parts Location, Front View



1 Capacitor C2
2 Resistors RD
3 #8 Terminal screws
4 Terminal 20
5 Terminal 19
6 Coil terminals (#8 screws)
7 Capacitor C1
8 Diode D
9 Resistor RF
10 #8 Terminal screws
11 Resistor RS

Figure 5-2. Contactor Parts Location, Left-hand Side View

Table 5-2. List of Parts Manufacturers

FSCM	Manufacturer
02989	GE Drive Systems 1501 Roanoke Blvd. Salem, VA 24153
44655	Ohmite Manufacturing Company xxxxxx xxxxxx
63743	Ward Leonard Electric Company xxxxxx xxxxxx

CHAPTER 6

INSTALLATION

6-1. INTRODUCTION

This chapter contains the information necessary to install the contactor. Before performing any installation work, consult with the drawings in this book.

6-2. RECEIVING AND HANDLING

General Electric carefully inspects and packs all equipment before shipping it from the factory. Upon receipt of this equipment, immediately inspect it for missing or damaged items. A packing list, included in each case of the equipment, itemizes the contents of each package of the shipment. When unpacking, check the contents of each case against the packing list. Carefully inspect the packing material to avoid losing small parts.

If damage is evident or if there is visual indication of rough handling, immediately file a damage claim with the carrier. Then notify both the transportation company and General Electric Company.

CAUTION

Before moving equipment, be sure that the doors are closed and their thumb screws are fastened.

6-3. UNPACKING

It is good practice to not completely unpack the equipment until it has been placed as near as possible to its permanent location. Use standard unpacking tools. Carefully remove equipment from its container to avoid damaging or marring the part.

Small parts (such as bolts and screws) are packed in special containers to keep them together, but they may become separated. For this reason, carefully inspect packing material for loose parts before discarding it. Wipe off any particles of packing material or foreign substances that may be lodged in or between the parts.

6-4. STORAGE

If the equipment is not installed immediately upon receipt, use the following storage procedures to protect it from breakage, corrosion, damage, and deterioration:

a. Unpack the equipment and label it.

b. The packing coverings do not protect the equipment for outdoor storage. Place the equipment under adequate cover, with the following protections:

(1) Keep the equipment clean and dry, and protect it from extreme temperature variations, high humidity, dust, and rodents.

(2) Use only breathable (canvas type) covering material – do not use plastic.

(3) Absolute storage temperature limits are between – 40 °C (– 40 °F) and 70 °C (158 °F).

(4) Equipment may be stored in environments of 5–95% relative humidity with provisions to prevent condensation

c. To prevent moisture condensation on the equipment, as caused by variations in storage temperature, keep the equipment's internal temperature 5–10 °C higher than outside ambient. This can be done by adding space heaters or by using panel space heaters, when supplied. (Heaters require an external ac power supply.)

CAUTION

Remove all cartons and other miscellaneous material packed inside units before energizing any heaters.

6-5. INSTALLATION

6-6. PREPARATION. Before starting any installation, consult and study all furnished drawings. These drawings should include arrangement drawings, connection diagrams, elementary diagrams, and a summary of the equipment.

6-7. MOUNTING. The contactor is to be mounted on a flat surface. The outline drawings (see figures 6-1 thru 6-3) specify the contactor size and the number, size, and location of bolts for mounting,

Mounting dimensions allow a normal factory tolerance of ± 1/16 inch. Therefore, alignment problems may occur if mounting provisions are based on drawing dimensions.

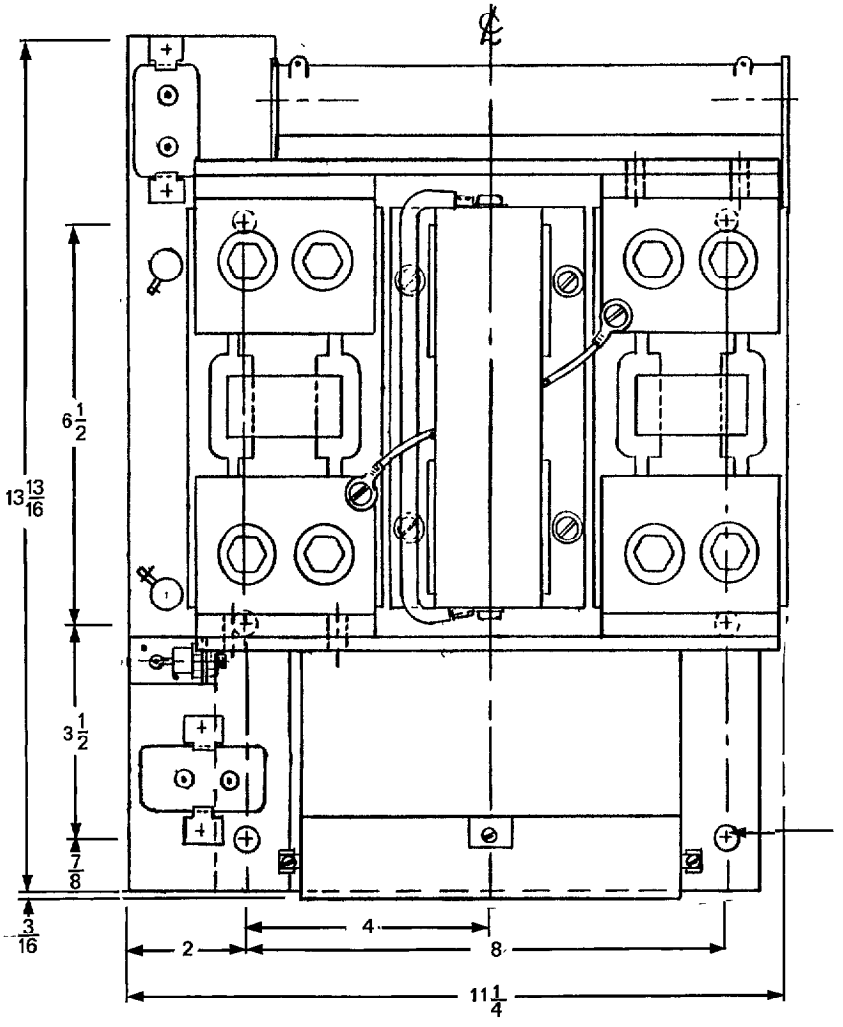


Figure 6-1. Contactor Outline and Mounting Details, Front View

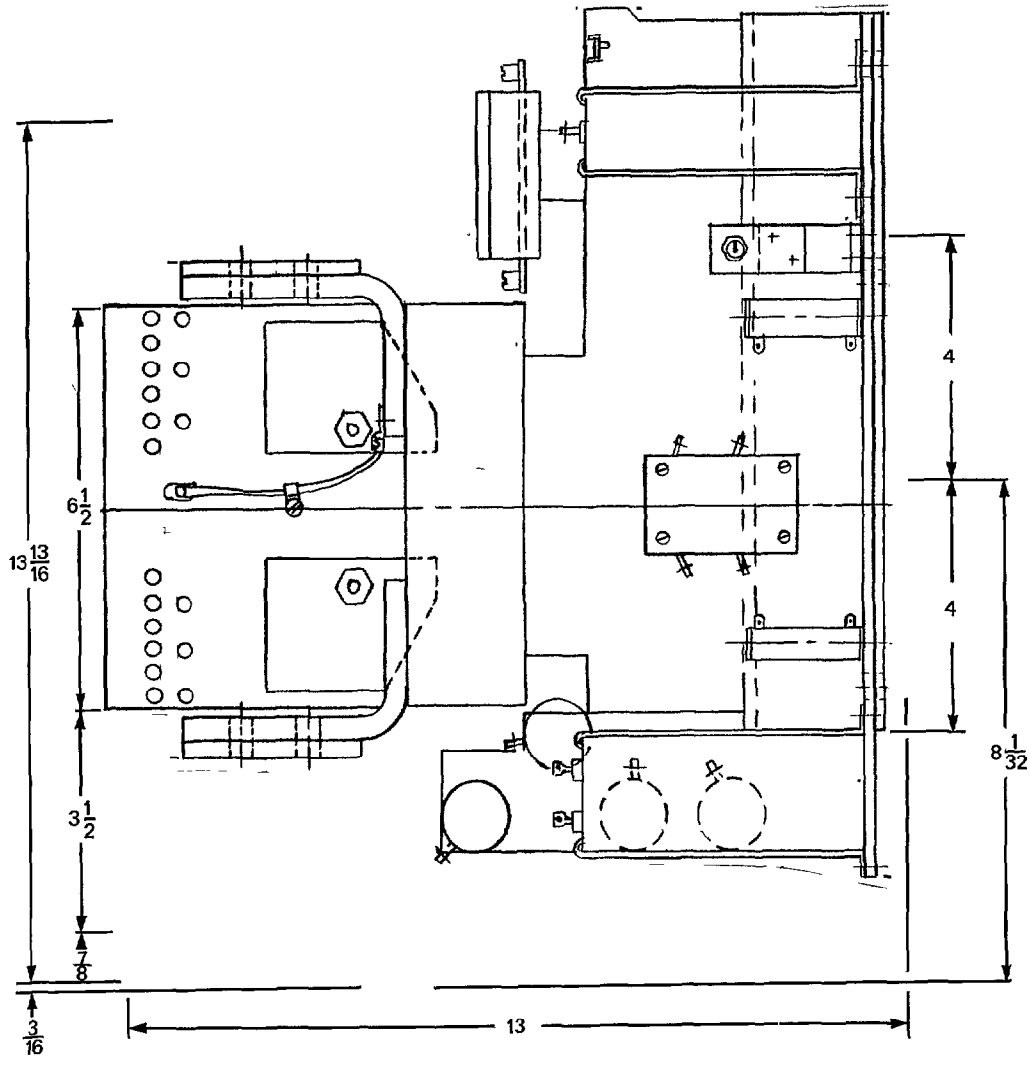


Figure 6-2. Contactor Outline and Mounting Details, Left-hand Side View

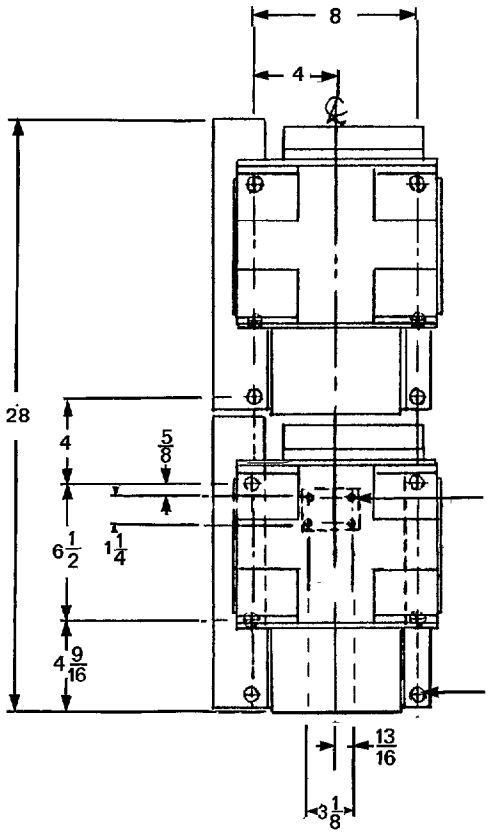


Figure 6-3. Contactor Outline and Mounting Details, Dimensions When Two Contactors Are Interlocked in Vertical Direction

6-8. INCOMING CABLING AND WIRING

WARNING

Before handling and connecting any wiring to the equipment, make sure that all power supplies are turned off. Then check voltage levels on the wiring to ensure that it is not carrying hazardous voltages. Tle 6-1 shows contactor wiring connections.

6-9. INSPECTION AND PREENERGIZING PROCEDURES

After completing installation, wiring, and normal circuit checks, the technician should complete the following actions before energizing the equipment:

a. Verify that each contactor is installed according to the specifications described.

b. Operate each magnetic device by hand to see that all moving parts operate freely. Check all electrical interlocks for proper operation.

c. If a malfunction is detected, troubleshoot according to the instructions in this manual.

Table 6-1. Wire Table
Refer to Figure 6-4

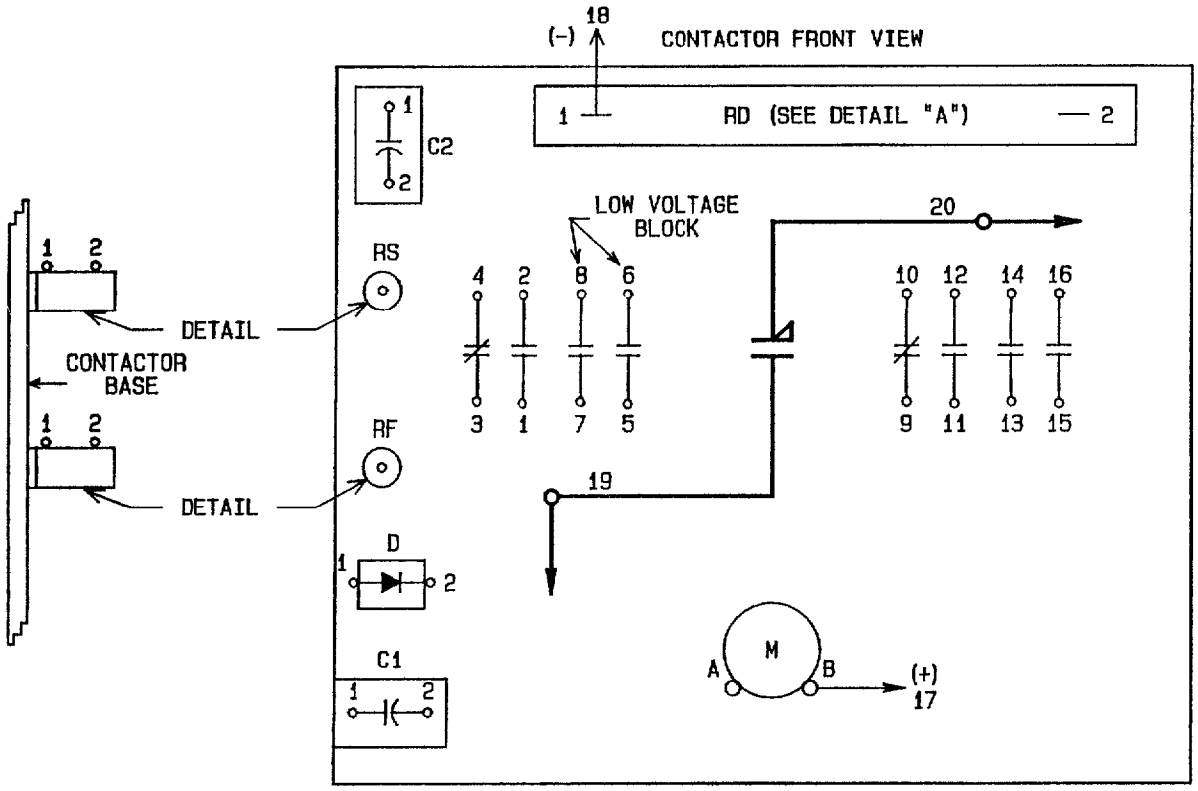


Figure 6-4. Contactor Wiring Connections